Selection on Observables I

Identification and Sub-Classification

PS200C, Spring 2024

Chad Hazlett

UCLA

chazlett@ucla.edu

Observational Studies

- Randomization allows unbiased estimation of $\mathbb{E}[Y_{0i}|D_i = 1]$ via $\mathbb{E}[Y_{0i}|D_i = 0]$ and $\mathbb{E}[Y_{1i}|D_i = 0]$ via $\mathbb{E}[Y_{1i}|D_i = 1]$.

Observational Studies

- Randomization allows unbiased estimation of $\mathbb{E}[Y_{0i}|D_i = 1]$ via $\mathbb{E}[Y_{0i}|D_i = 0]$ and $\mathbb{E}[Y_{1i}|D_i = 0]$ via $\mathbb{E}[Y_{1i}|D_i = 1]$.

  Equivalently: We can think about influences on POs rather than the POs, with randomzation "balancing" observed and unobserved influences on POs

Observational Studies

- Randomization allows unbiased estimation of $\mathbb{E}[Y_{0i}|D_i = 1]$ via $\mathbb{E}[Y_{0i}|D_i = 0]$ and $\mathbb{E}[Y_{1i}|D_i = 0]$ via $\mathbb{E}[Y_{1i}|D_i = 1]$.

  Equivalently: We can think about influences on POs rather than the POs, with randomzation "balancing" observed and unobserved influences on POs

- When we cannot randomize, we do observational studies, primarily meaning we adjust for observed covariates and hope that unobservables are balanced

Observational Studies

- Randomization allows unbiased estimation of $\mathbb{E}[Y_{0i}|D_i = 1]$ via $\mathbb{E}[Y_{0i}|D_i = 0]$ and $\mathbb{E}[Y_{1i}|D_i = 0]$ via $\mathbb{E}[Y_{1i}|D_i = 1]$.

  Equivalently: We can think about influences on POs rather than the POs, with randomzation "balancing" observed and unobserved influences on POs

- When we cannot randomize, we do observational studies, primarily meaning we adjust for observed covariates and hope that unobservables are balanced

Most of the rest of the course: what to do in observational settings.

Observational Studies

- Randomization allows unbiased estimation of $\mathbb{E}[Y_{0i}|D_i = 1]$ via $\mathbb{E}[Y_{0i}|D_i = 0]$ and $\mathbb{E}[Y_{1i}|D_i = 0]$ via $\mathbb{E}[Y_{1i}|D_i = 1]$.

  Equivalently: We can think about influences on POs rather than the POs, with randomzation "balancing" observed and unobserved influences on POs

- When we cannot randomize, we do observational studies, primarily meaning we adjust for observed covariates and hope that unobservables are balanced

Most of the rest of the course: what to do in observational settings.

- What is your strategy for identifying the missing potential outcomes from observational data?
- What assumptions does it involve? Can we weaken these?
- Are they credible? What can you test? What do you need to argue?

Good Observational Studies

Preview: Some designs we will introduce:

- "Finding experiments" in the data: constructing comparison groups s.t. treatment assignment arguably haphazard. Keywords: conditional ignorability, controlling for common causes, no unobserved confounders, selection on observables.

Good Observational Studies

Preview: Some designs we will introduce:

- "Finding experiments" in the data: constructing comparison groups s.t. treatment assignment arguably haphazard. Keywords: conditional ignorability, controlling for common causes, no unobserved confounders, selection on observables.
- Over-time comparisons to remove time-invariant confounders. Keywords: Fixed effects

Good Observational Studies

Preview: Some designs we will introduce:

- "Finding experiments" in the data: constructing comparison groups s.t. treatment assignment arguably haphazard. Keywords: conditional ignorability, controlling for common causes, no unobserved confounders, selection on observables.
- Over-time comparisons to remove time-invariant confounders. Keywords: Fixed effects
- Over-time and between group comparisons to remove time-invariant effects and effects of time. Keywords: Diff-in-Diff, panel, synthetic control

Good Observational Studies

Preview: Some designs we will introduce:

- "Finding experiments" in the data: constructing comparison groups s.t. treatment assignment arguably haphazard. Keywords: conditional ignorability, controlling for common causes, no unobserved confounders, selection on observables.
- Over-time comparisons to remove time-invariant confounders. Keywords: Fixed effects
- Over-time and between group comparisons to remove time-invariant effects and effects of time. Keywords: Diff-in-Diff, panel, synthetic control
- Finding surrogates for treatment or "the random component" of treatment Keywords: Instrumental variables.

Good Observational Studies

Preview: Some designs we will introduce:

- "Finding experiments" in the data: constructing comparison groups s.t. treatment assignment arguably haphazard. Keywords: conditional ignorability, controlling for common causes, no unobserved confounders, selection on observables.
- Over-time comparisons to remove time-invariant confounders. Keywords: Fixed effects
- Over-time and between group comparisons to remove time-invariant effects and effects of time. Keywords: Diff-in-Diff, panel, synthetic control
- Finding surrogates for treatment or "the random component" of treatment Keywords: Instrumental variables.
- Leveraging sudden changes. Keywords: Regression discontinuity, interupted time-series.

Good Observational Studies

Preview: Some designs we will introduce:

- "Finding experiments" in the data: constructing comparison groups s.t. treatment assignment arguably haphazard. Keywords: conditional ignorability, controlling for common causes, no unobserved confounders, selection on observables.
- Over-time comparisons to remove time-invariant confounders. Keywords: Fixed effects
- Over-time and between group comparisons to remove time-invariant effects and effects of time. Keywords: Diff-in-Diff, panel, synthetic control
- Finding surrogates for treatment or "the random component" of treatment Keywords: Instrumental variables.
- Leveraging sudden changes. Keywords: Regression discontinuity, interupted time-series.
- Leveraging over time changes in treatment availability, while allowing selection-into-treatment. Keywords: Stability controlled quasi-experiment.
- Allow violated assumptions but clarify how violated they would have to be to alter conclusion. Keywords: Sensitivity analyses.

The First Commandment of Causal Inference

A result is causal if you can support the assumptions that make it causal, not because you performed some procedure.

- talking about and learning estimation tools is great but no estimation tool **makes** your result causal; rather the assumptions tell you what you'd have to believe to believe the result is causal.
- under a given set of assumptions, a variety of estimators may be suitable and the choice may matter less than the credibility rooted in the assumptions.
- most of your effort should go into finding an ID strategy with the most suitable assumptions for your case, then understanding and communicating whether the assumptions are credible or how violations may have changed conclusion.

Corollary to First Commandment:

Never say "my estimate is causal because I used [estimator]."

(Where [estimator]=matching, IV, whatever.)

Outline

1 Our first non-random ID Strategy: CI/SOO

2 Estimation by Subclassification

Setting

- Units: $i = 1, ..., n$
- Treatment: $D_i \in \{0, 1\}$
- Potential outcomes: Y_{di}, where $d = \{0, 1\}$

Setting

- Units: $i = 1, ..., n$
- Treatment: $D_i \in \{0, 1\}$
- Potential outcomes: Y_{di}, where $d = \{0, 1\}$
- Quantities of interest:

$$\begin{aligned} \text{ATE:} \quad \tau_{ATE} &\equiv \mathbb{E}[Y_{1i} - Y_{0i}] \\ \text{ATT:} \quad \tau_{ATT} &\equiv \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1] \end{aligned}$$

Setting

- Units: $i = 1, ..., n$
- Treatment: $D_i \in \{0, 1\}$
- Potential outcomes: Y_{di}, where $d = \{0, 1\}$
- Quantities of interest:

$$\begin{aligned} \text{ATE:} \quad \tau_{ATE} &\equiv \mathbb{E}[Y_{1i} - Y_{0i}] \\ \text{ATT:} \quad \tau_{ATT} &\equiv \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1] \end{aligned}$$

- Pre-treatment covariates: $X_i = [X_{i1}, ..., X_{iK}]^\top \in \mathcal{X}$
  - Examples: Sex, race, age, etc.
  - X_i may be correlated with both D_i and Y_{di}, thereby confounding the causal relationship
  - For now, take care that the covariates are not affected by D_i, but there are some other trouble cases we will get to with DAGs.

Conditional Ignorability/ Selection on Observables

Recall that randomized experiments work because:

$$\{Y_{0i}, Y_{1i}\} \perp\!\!\!\perp D_i$$

Conditional Ignorability/ Selection on Observables

Recall that randomized experiments work because:

$$\{Y_{0i}, Y_{1i}\} \perp\!\!\!\perp D_i$$

Let's change it a bit:

Assumption: Conditional Ignorability

$$\{Y_{0i}, Y_{1i}\} \perp\!\!\!\perp D_i \mid X_i = x \quad \text{for any} \quad x \in \mathcal{X}$$

This goes by many names: strong ignorability, unconfoundedness, selection on observables, conditionally as-if random, no omitted variables

Conditional Ignorability/ Selection on Observables

Recall that randomized experiments work because:

$$\{Y_{0i}, Y_{1i}\} \perp\!\!\!\perp D_i$$

Let's change it a bit:

Assumption: Conditional Ignorability

$$\{Y_{0i}, Y_{1i}\} \perp\!\!\!\perp D_i \mid X_i = x \quad \text{for any} \quad x \in \mathcal{X}$$

This goes by many names: strong ignorability, unconfoundedness, selection on observables, conditionally as-if random, no omitted variables

Tip: Read it backwards..."'Among units with same X, D_i is (as good as) randomly assigned."

Conditional Ignorability/ Selection on Observables

What does it really mean?

Assumption: Conditional Ignorability

$$\{Y_{0i}, Y_{1i}\} \perp\!\!\!\perp D_i \mid X_i = x \quad \text{for any} \quad x \in \mathcal{X}$$

This *would be satisfied by design* if you performed "conditional experiments": imagine taking units with some value of X and conducting a randomized experiment with them ($Pr(D)$ can vary by group).

Conditional Ignorability/ Selection on Observables

What does it really mean?

Assumption: Conditional Ignorability

$$\{Y_{0i}, Y_{1i}\} \perp\!\!\!\perp D_i \mid X_i = x \quad \text{for any} \quad x \in \mathcal{X}$$

This *would be satisfied by design* if you performed "conditional experiments": imagine taking units with some value of X and conducting a randomized experiment with them ($Pr(D)$ can vary by group).

- While one could run an experiment this way, this will get applied to observational studies where nobody designed the treatment assignment this way.

Conditional Ignorability/ Selection on Observables

What does it really mean?

Assumption: Conditional Ignorability

$$\{Y_{0i}, Y_{1i}\} \perp\!\!\!\perp D_i \mid X_i = x \quad \text{for any} \quad x \in \mathcal{X}$$

This *would be satisfied by design* if you performed "conditional experiments": imagine taking units with some value of X and conducting a randomized experiment with them ($Pr(D)$ can vary by group).

- While one could run an experiment this way, this will get applied to observational studies where nobody designed the treatment assignment this way.
- What *could* make this credible in an observational study?
  - Some natural, accidental treatment assignment resembling the "conditionally randomized experiment"
  - You understand factors influencing treatment so that only "noise" can remain.
  - Any other way you can argue there are "no unobserved confounders"

Conditional Ignorability/ Selection on Observables

What does it really mean?

Assumption: Conditional Ignorability

$$\{Y_{0i}, Y_{1i}\} \perp\!\!\!\perp D_i \mid X_i = x \quad \text{for any} \quad x \in \mathcal{X}$$

This *would be satisfied by design* if you performed "conditional experiments": imagine taking units with some value of X and conducting a randomized experiment with them ($Pr(D)$ can vary by group).

- While one could run an experiment this way, this will get applied to observational studies where nobody designed the treatment assignment this way.
- What *could* make this credible in an observational study?
  - Some natural, accidental treatment assignment resembling the "conditionally randomized experiment"
  - You understand factors influencing treatment so that only "noise" can remain.
  - Any other way you can argue there are "no unobserved confounders"
- Most observational work is conducted **as though this was exactly true**.

Conditional Ignorability/ Selection on Observables

What does it really mean?

Assumption: Conditional Ignorability

$$\{Y_{0i}, Y_{1i}\} \perp\!\!\!\perp D_i \mid X_i = x \quad \text{for any} \quad x \in \mathcal{X}$$

This *would be satisfied by design* if you performed "conditional experiments": imagine taking units with some value of X and conducting a randomized experiment with them ($Pr(D)$ can vary by group).

- While one could run an experiment this way, this will get applied to observational studies where nobody designed the treatment assignment this way.
- What *could* make this credible in an observational study?
  - Some natural, accidental treatment assignment resembling the "conditionally randomized experiment"
  - You understand factors influencing treatment so that only "noise" can remain.
  - Any other way you can argue there are "no unobserved confounders"
- Most observational work is conducted **as though this was exactly true**.
- The threat that this is wrong, and the bias that can result, is **not alleviated by using more sophisticated looking estimators, or larger sample sizes**.

Common Support

Another assumption we'll often need to use:

Common Support

Another assumption we'll often need to use:

Assumption: Common Support

$$0 < \Pr(D_i = 1 \mid X_i = x) < 1 \quad \text{for any} \quad x \in \mathcal{X}$$

Common Support

Another assumption we'll often need to use:

Assumption: Common Support

$$0 < \Pr(D_i = 1 \mid X_i = x) < 1 \quad \text{for any} \quad x \in \mathcal{X}$$

Read:

Common Support

Another assumption we'll often need to use:

Assumption: Common Support

$$0 < \Pr(D_i = 1 \mid X_i = x) < 1 \quad \text{for any} \quad x \in \mathcal{X}$$

Read: For any value of X_i, unit could have received treatment or control

ATE, ATT, ATC Identified under SOO

Intuition: Within strata of X, you have an experiment, and you just need to recombine these.

ATE, ATT, ATC Identified under SOO

Intuition: Within strata of X, you have an experiment, and you just need to recombine these.

Proof: for ATE, τ, we have

ATE, ATT, ATC Identified under SOO

Intuition: Within strata of X, you have an experiment, and you just need to recombine these.

Proof: for ATE, τ, we have

Part 1. Identifiability of $\tau(X)$:

$$\begin{aligned}\mathbb{E}[Y_{1i} - Y_{0i}|X_i = x] &= \mathbb{E}[Y_{1i}|X_i = x] - \mathbb{E}[Y_{0i}|X_i = x] \\ &= \mathbb{E}[Y_{1i}|X_i = x, D_i = 1] - \mathbb{E}[Y_{0i}|X_i = x, D_i = 0] \\ &= \mathbb{E}[Y_i|X_i = x, D_i = 1] - \mathbb{E}[Y_i|X_i = x, D_i = 0]\end{aligned}$$

ATE, ATT, ATC Identified under SOO

Intuition: Within strata of X, you have an experiment, and you just need to recombine these.

Proof: for ATE, τ, we have

Part 1. Identifiability of $\tau(X)$:

$$\begin{aligned}\mathbb{E}[Y_{1i} - Y_{0i}|X_i = x] &= \mathbb{E}[Y_{1i}|X_i = x] - \mathbb{E}[Y_{0i}|X_i = x] \\ &= \mathbb{E}[Y_{1i}|X_i = x, D_i = 1] - \mathbb{E}[Y_{0i}|X_i = x, D_i = 0] \\ &= \mathbb{E}[Y_i|X_i = x, D_i = 1] - \mathbb{E}[Y_i|X_i = x, D_i = 0]\end{aligned}$$

Part 2. Common support gets you back to τ:

$$\begin{aligned}\tau_{ATE} &= \mathbb{E}[Y_{1i} - Y_{0i}] \\ &= \mathbb{E}\big[\mathbb{E}[Y_{1i} - Y_{0i}|X_i]\big] \\ &= \int \Big(\mathbb{E}[Y_i|D_i = 1, X_i = x] - \mathbb{E}[Y_i|D_i = 0, X_i = x]\Big)p(x)dX \\ &= \mathbb{E}\big[\mathbb{E}[\hat{\tau}|X_i]\big] = \mathbb{E}[\hat{\tau}]\end{aligned}$$

Identification of ATT

By similar logic, τ_{ATT} is also identified under the conditional ignorability and common support assumptions:

$$\tau_{ATT} = \mathbb{E}[\tau(X_i) \mid D_i = 1]$$

Identification of ATT

By similar logic, τ_{ATT} is also identified under the conditional ignorability and common support assumptions:

$$\tau_{ATT} = \mathbb{E}[\tau(X_i) \mid D_i = 1]$$

Proof is similar:

$$\begin{aligned}\tau_{ATT} &= \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1] \\ &= \mathbb{E}_x[\mathbb{E}[Y_{1i} - Y_{0i} \mid X_i, D_i = 1]] \\ &= \mathbb{E}_x[\mathbb{E}[Y_i|X_i, D_i = 1] - \mathbb{E}[Y_i|X_i, D_i = 0]]\end{aligned}$$

Identification of ATT

By similar logic, τ_{ATT} is also identified under the conditional ignorability and common support assumptions:

$$\tau_{ATT} = \mathbb{E}[\tau(X_i) \mid D_i = 1]$$

Proof is similar:

$$\begin{aligned}\tau_{ATT} &= \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1] \\ &= \mathbb{E}_x[\mathbb{E}[Y_{1i} - Y_{0i} \mid X_i, D_i = 1]] \\ &= \mathbb{E}_x[\mathbb{E}[Y_i|X_i, D_i = 1] - \mathbb{E}[Y_i|X_i, D_i = 0]]\end{aligned}$$

Is $\tau_{ATE} = \tau_{ATT}$ when SOO holds?

Why don't we condition on post-treatment variables?

We will revisit this with DAGs, but here are some intuitions.

Why don't we condition on post-treatment variables?

We will revisit this with DAGs, but here are some intuitions.

At minimum, you might eat up part of the effect of the treatment

Why don't we condition on post-treatment variables?

We will revisit this with DAGs, but here are some intuitions.

At minimum, you might eat up part of the effect of the treatment

While that might be acceptable for some estimands, absent other strong assumptions, it does more damage than that.

Why don't we condition on post-treatment variables?

We will revisit this with DAGs, but here are some intuitions.

At minimum, you might eat up part of the effect of the treatment

While that might be acceptable for some estimands, absent other strong assumptions, it does more damage than that.

Let's consider a concrete example:

- suppose you randomly assign towns to community policing or nothing ($D = 1$ or $D = 0$), check whether there are community discussions of policing after that ($Z = 1$ or $Z = 0$), and measure the crime rate (Y).

Why don't we condition on post-treatment variables?

We will revisit this with DAGs, but here are some intuitions.

At minimum, you might eat up part of the effect of the treatment

While that might be acceptable for some estimands, absent other strong assumptions, it does more damage than that.

Let's consider a concrete example:

- suppose you randomly assign towns to community policing or nothing ($D = 1$ or $D = 0$), check whether there are community discussions of policing after that ($Z = 1$ or $Z = 0$), and measure the crime rate (Y).
- sadly you find no effect of D on Y, then you wonder, maybe it only worked where the community actually engaged? ($Z = 1$).

Why don't we condition on post-treatment variables?

We will revisit this with DAGs, but here are some intuitions.

At minimum, you might eat up part of the effect of the treatment

While that might be acceptable for some estimands, absent other strong assumptions, it does more damage than that.

Let's consider a concrete example:

- suppose you randomly assign towns to community policing or nothing ($D = 1$ or $D = 0$), check whether there are community discussions of policing after that ($Z = 1$ or $Z = 0$), and measure the crime rate (Y).
- sadly you find no effect of D on Y, then you wonder, maybe it only worked where the community actually engaged? ($Z = 1$).

So compare treated to control but only among those with $Z = 1$, right?

Why don't we condition on post-treatment variables?

We will revisit this with DAGs, but here are some intuitions.

At minimum, you might eat up part of the effect of the treatment

While that might be acceptable for some estimands, absent other strong assumptions, it does more damage than that.

Let's consider a concrete example:

- suppose you randomly assign towns to community policing or nothing ($D = 1$ or $D = 0$), check whether there are community discussions of policing after that ($Z = 1$ or $Z = 0$), and measure the crime rate (Y).
- sadly you find no effect of D on Y, then you wonder, maybe it only worked where the community actually engaged? ($Z = 1$).

So compare treated to control but only among those with $Z = 1$, right?

- Noooo.....but why?

Why don't we condition on post-treatment variables?

Here is a particular way the world might work, but the concern generalizes:

Why don't we condition on post-treatment variables?

Here is a particular way the world might work, but the concern generalizes:

- Suppose that to get $Z = 1$ you need either treatment, or high social capital (or both)

Why don't we condition on post-treatment variables?

Here is a particular way the world might work, but the concern generalizes:

- Suppose that to get $Z = 1$ you need either treatment, or high social capital (or both)
- Among the treated, those with $Z = 1$ will have a natural mix of high and low social capital

Why don't we condition on post-treatment variables?

Here is a particular way the world might work, but the concern generalizes:

- Suppose that to get $Z = 1$ you need either treatment, or high social capital (or both)
- Among the treated, those with $Z = 1$ will have a natural mix of high and low social capital
- But among the controls, to look only at those with $Z = 1$ means dropping the units with low social capital...social capital will be higher.

Why don't we condition on post-treatment variables?

Here is a particular way the world might work, but the concern generalizes:

- Suppose that to get $Z = 1$ you need either treatment, or high social capital (or both)
- Among the treated, those with $Z = 1$ will have a natural mix of high and low social capital
- But among the controls, to look only at those with $Z = 1$ means dropping the units with low social capital...social capital will be higher.
- Hence the treated and control groups, conditional on $Z = 1$, now differ on social capital

Why don't we condition on post-treatment variables?

Here is a particular way the world might work, but the concern generalizes:

- Suppose that to get $Z = 1$ you need either treatment, or high social capital (or both)
- Among the treated, those with $Z = 1$ will have a natural mix of high and low social capital
- But among the controls, to look only at those with $Z = 1$ means dropping the units with low social capital...social capital will be higher.
- Hence the treated and control groups, conditional on $Z = 1$, now differ on social capital

The heuristic thing to remember: don't condition/stratify on post-treatment variables!

Estimation

We know what assumptions would allow us to *identified* causal effects under conditional ignorability.

Two challenges remain:

1. How do we evaluate the credibility of the SOO/CI assumptions?
2. How do we actually do the conditioning needed for estimation under SOO/CI?

Estimation

We know what assumptions would allow us to *identified* causal effects under conditional ignorability.

Two challenges remain:

1. How do we evaluate the credibility of the SOO/CI assumptions?
2. How do we actually do the conditioning needed for estimation under SOO/CI?

These two questions are ubiquitous in causal inference work. We will spend the next few sessions alternating between these.

Estimation

We know what assumptions would allow us to *identified* causal effects under conditional ignorability.

Two challenges remain:

1. How do we evaluate the credibility of the SOO/CI assumptions?
2. How do we actually do the conditioning needed for estimation under SOO/CI?

These two questions are ubiquitous in causal inference work. We will spend the next few sessions alternating between these.

Estimation

We know what assumptions would allow us to *identified* causal effects under conditional ignorability.

Two challenges remain:

1. How do we evaluate the credibility of the SOO/CI assumptions?
2. How do we actually do the conditioning needed for estimation under SOO/CI?

These two questions are ubiquitous in causal inference work. We will spend the next few sessions alternating between these.

For estimation under SOO, we will explore:

- sub-classification
- matching: with and without propensity score
- re-weighting: with and without propensity score
- regression, model-based imputation

Estimation

We know what assumptions would allow us to *identified* causal effects under conditional ignorability.

Two challenges remain:

1. How do we evaluate the credibility of the SOO/CI assumptions?
2. How do we actually do the conditioning needed for estimation under SOO/CI?

These two questions are ubiquitous in causal inference work. We will spend the next few sessions alternating between these.

For estimation under SOO, we will explore:

- sub-classification
- matching: with and without propensity score
- re-weighting: with and without propensity score
- regression, model-based imputation

Today we start with the most natural, sub-classification

Outline

1 Our first non-random ID Strategy: CI/SOO

2 Estimation by Subclassification

Identification Results for Discrete Covariates

Sub-classification is just "actually conditioning on X", i.e. stratifying.

Identification Results for Discrete Covariates

Sub-classification is just "actually conditioning on X", i.e. stratifying.

If X_i is discrete, we can literally use our identification results:

$$\tau_{ATE} = \sum_{x \in \mathcal{X}} \{\mathbb{E}[Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[Y_i \mid D_i = 0, X_i = x]\} \Pr(X_i = x)$$

Identification Results for Discrete Covariates

Sub-classification is just "actually conditioning on X", i.e. stratifying.

If X_i is discrete, we can literally use our identification results:

$$\tau_{ATE} = \sum_{x \in \mathcal{X}} \{\mathbb{E}[Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[Y_i \mid D_i = 0, X_i = x]\} \Pr(X_i = x)$$

The sample analog estimator for this will be:

$$\hat{\tau}_{ATE} = \sum_{x \in \mathcal{X}} \left\{\hat{\mathbb{E}}[Y_i \mid D_i = 1, X_i = x] - \hat{\mathbb{E}}[Y_i \mid D_i = 0, X_i = x]\right\} \hat{\Pr}(X_i = x)$$

where $\hat{\mathbb{E}}[\cdot]$ is my lazy way to write a sample average.

Identification Results for Discrete Covariates

Sub-classification is just "actually conditioning on X", i.e. stratifying.

If X_i is discrete, we can literally use our identification results:

$$\tau_{ATE} = \sum_{x \in \mathcal{X}} \{\mathbb{E}[Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[Y_i \mid D_i = 0, X_i = x]\} \Pr(X_i = x)$$

The sample analog estimator for this will be:

$$\hat{\tau}_{ATE} = \sum_{x \in \mathcal{X}} \left\{\hat{\mathbb{E}}[Y_i \mid D_i = 1, X_i = x] - \hat{\mathbb{E}}[Y_i \mid D_i = 0, X_i = x]\right\} \hat{\Pr}(X_i = x)$$

where $\hat{\mathbb{E}}[\cdot]$ is my lazy way to write a sample average.

In other words: $\hat{\tau}_{ATE}$ can be calculated by:

(1) Group units into strata defined by the values of X_i.

(2) For each stratum, calculate difference in means of Y_i between the treated and untreated.

(3) Calculate the weighted average of (2), with weights equal to the proportions of units in the strata.

τ_{ATT} can be calculated similarly:

$$\tau_{ATT} = \sum_{x \in \mathcal{X}} \{\mathbb{E}[Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[Y_i \mid D_i = 0, X_i = x]\} \Pr(X_i = x \mid D_i = 1)$$

τ_{ATT} can be calculated similarly:

$$\tau_{ATT} = \sum_{x \in \mathcal{X}} \{\mathbb{E}[Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[Y_i \mid D_i = 0, X_i = x]\} \Pr(X_i = x \mid D_i = 1)$$

With sample analog estimator:

$$\hat{\tau}_{ATT} = \sum_{x \in \mathcal{X}} \left\{\hat{\mathbb{E}}[Y_i \mid D_i = 1, X_i = x] - \hat{\mathbb{E}}[Y_i \mid D_i = 0, X_i = x]\right\} \hat{\Pr}(X_i = x | D_i = 1)$$

τ_{ATT} can be calculated similarly:

$$\tau_{ATT} = \sum_{x \in \mathcal{X}} \{\mathbb{E}[Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[Y_i \mid D_i = 0, X_i = x]\} \Pr(X_i = x \mid D_i = 1)$$

With sample analog estimator:

$$\hat{\tau}_{ATT} = \sum_{x \in \mathcal{X}} \left\{\hat{\mathbb{E}}[Y_i \mid D_i = 1, X_i = x] - \hat{\mathbb{E}}[Y_i \mid D_i = 0, X_i = x]\right\} \hat{\Pr}(X_i = x | D_i = 1)$$

In other words: $\hat{\tau}_{ATE}$ can be calculated by:

(1) Group units into strata defined by the values of X_i.

(2) For each stratum, calculate difference in means of Y_i between the treated and untreated.

(3) Calculate the weighted average of (2), with weights equal to the proportions of units in the strata within the treatment group.

Subclassification Estimators

This gives us the subclassification estimators:

$$\hat{\tau}_{ATE} = \sum_{j=1}^{M} \left\{ \overline{Y}_{1j} - \overline{Y}_{0j} \right\} \frac{n_j}{n}$$

$$\hat{\tau}_{ATT} = \sum_{j=1}^{M} \left\{ \overline{Y}_{1j} - \overline{Y}_{0j} \right\} \frac{n_{1j}}{n_1}$$

where
$$\begin{cases} M & = & \text{\# of strata} \\ n_j & = & \text{\# of units in cell } j \\ n_{1j} & = & \text{\# of treated units in cell } j \\ \overline{Y}_{dj} & = & \text{mean outcome for units with } D_i = d \text{ in cell } j \end{cases}$$

Example: Smoking and Mortality (Cochran 1968)

TABLE 1

DEATH RATES PER 1,000 PERSON-YEARS

Smoking group	Canada	U.K.	U.S.
Non-smokers	20.2	11.3	13.5
Cigarettes	20.5	14.1	13.5
Cigars/pipes	35.5	20.7	17.4

Example: Smoking and Mortality (Cochran 1968)

TABLE 2

MEAN AGES, YEARS

Smoking group	Canada	U.K.	U.S.
Non-smokers	54.9	49.1	57.0
Cigarettes	50.5	49.8	53.2
Cigars/pipes	65.9	55.7	59.7

Subclassification: Example

	Death Rates Pipe Smokers	# Pipe-Smokers	# Non-Smokers
Age 20 - 50	15	11	29
Age 50 - 70	35	13	9
Age + 70	50	16	2
Total		40	40

What is the average death rate for Pipe Smokers?

Subclassification: Example

	Death Rates Pipe Smokers	# Pipe-Smokers	# Non-Smokers
Age 20 - 50	15	11	29
Age 50 - 70	35	13	9
Age + 70	50	16	2
Total		40	40

What is the average death rate for Pipe Smokers?

$15 \cdot (11/40) + 35 \cdot (13/40) + 50 \cdot (16/40) = 35.5$

Subclassification: Example

	Death Rates Pipe Smokers	# Pipe-Smokers	# Non-Smokers
Age 20 - 50	15	11	29
Age 50 - 70	35	13	9
Age + 70	50	16	2
Total		40	40

What would be the average death rate for Pipe Smokers if they had the same age distribution as Non-Smokers?

Subclassification: Example

	Death Rates Pipe Smokers	# Pipe-Smokers	# Non-Smokers
Age 20 - 50	15	11	29
Age 50 - 70	35	13	9
Age + 70	50	16	2
Total		40	40

What would be the average death rate for Pipe Smokers if they had the same age distribution as Non-Smokers?

$15 \cdot (29/40) + 35 \cdot (9/40) + 50 \cdot (2/40) = 21.2$

Smoking and Mortality (Cochran (1968))

TABLE 3

ADJUSTED DEATH RATES USING 3 AGE GROUPS

Smoking group	Canada	U.K.	U.S.
Non-smokers	20.2	11.3	13.5
Cigarettes	28.3	12.8	17.7
Cigars/pipes	21.2	12.0	14.2

Subclassification by Age ($M = 2$)

X_j	Death Rate Smokers	Death Rate Non-Smokers	# Smokers	# Obs.
Old	28	24	3	10
Young	22	16	7	10
Total			10	20

Subclassification by Age ($M = 2$)

X_j	Death Rate Smokers	Death Rate Non-Smokers	# Smokers	# Obs.
Old	28	24	3	10
Young	22	16	7	10
Total			10	20

What would the (unadjusted) difference in mean death rates for smokers versus non-smokers be?

Subclassification by Age ($M = 2$)

X_j	Death Rate Smokers	Death Rate Non-Smokers	# Smokers	# Obs.
Old	28	24	3	10
Young	22	16	7	10
Total			10	20

What would the (unadjusted) difference in mean death rates for smokers versus non-smokers be?

$$[28(3/10) + 22(7/10)] - [24(7/10) + 16(3/10)]$$
$$= 2.2$$

Subclassification by Age ($M = 2$)

X_j	Death Rate Smokers	Death Rate Non-Smokers	# Smokers	# Obs.
Old	28	24	3	10
Young	22	16	7	10
Total			10	20

What would the (unadjusted) difference in mean death rates for smokers versus non-smokers be?

$$[28(3/10) + 22(7/10)] - [24(7/10) + 16(3/10)]$$
$$= 2.2$$

What is the subclassification estimator for the ATE of smoking?

Subclassification by Age ($M = 2$)

X_j	Death Rate Smokers	Death Rate Non-Smokers	# Smokers	# Obs.
Old	28	24	3	10
Young	22	16	7	10
Total			10	20

What would the (unadjusted) difference in mean death rates for smokers versus non-smokers be?

$$[28(3/10) + 22(7/10)] - [24(7/10) + 16(3/10)]$$
$$= 2.2$$

What is the subclassification estimator for the ATE of smoking?

$$\hat{\tau}_{ATE} = (28 - 24) \cdot \frac{10}{20} + (22 - 16) \cdot \frac{10}{20} = 5$$

Subclassification by Age and Gender ($M = 4$)

X_j	Death Rate Smokers	Death Rate Non-Smokers	# Smokers	# Obs.
Old, Male	28	22	3	7
Old, Female		24	0	3
Young, Male	21	16	3	4
Young, Female	23	17	4	6
Total			10	20

What is the subclassification estimate for the ATE of smoking on death rate?

Subclassification by Age and Gender ($M = 4$)

X_j	Death Rate Smokers	Death Rate Non-Smokers	# Smokers	# Obs.
Old, Male	28	22	3	7
Old, Female		24	0	3
Young, Male	21	16	3	4
Young, Female	23	17	4	6
Total			10	20

What is the subclassification estimate for the ATE of smoking on death rate?

Not identified! (because of the lack of common support)

Subclassification by Age and Gender ($M = 4$)

X_j	Death Rate Smokers	Death Rate Non-Smokers	# Smokers	# Obs.
Old, Male	28	22	3	7
Old, Female		24	0	3
Young, Male	21	16	3	4
Young, Female	23	17	4	6
Total			10	20

What is the subclassification estimate for the ATT of smoking on death rate?

Subclassification by Age and Gender ($M = 4$)

X_j	Death Rate Smokers	Death Rate Non-Smokers	# Smokers	# Obs.
Old, Male	28	22	3	7
Old, Female		24	0	3
Young, Male	21	16	3	4
Young, Female	23	17	4	6
Total			10	20

What is the subclassification estimate for the ATT of smoking on death rate?

$$\begin{aligned}\hat{\tau}_{ATT} &= (28-22)\cdot\frac{3}{10}+(21-16)\cdot\frac{3}{10}+(23-17)\cdot\frac{4}{10}\\ &= 5.1\end{aligned}$$

The most important slide

What was the first commandment? How does it apply here?

The most important slide

What was the first commandment? How does it apply here?

And what are the assumptions we need to get a causal effect estimate of the type of smoking on death rate?

The most important slide

What was the first commandment? How does it apply here?

And what are the assumptions we need to get a causal effect estimate of the type of smoking on death rate?

- Let's state it both in a sentence about confounding, and using the CI formalization.

The most important slide

What was the first commandment? How does it apply here?

And what are the assumptions we need to get a causal effect estimate of the type of smoking on death rate?

- Let's state it both in a sentence about confounding, and using the CI formalization.
- While we are at it, let's draw a DAG too.

The most important slide

What was the first commandment? How does it apply here?

And what are the assumptions we need to get a causal effect estimate of the type of smoking on death rate?

- Let's state it both in a sentence about confounding, and using the CI formalization.

- While we are at it, let's draw a DAG too.
- Do you believe this? What's a counter-example?

The most important slide

What was the first commandment? How does it apply here?

And what are the assumptions we need to get a causal effect estimate of the type of smoking on death rate?

- Let's state it both in a sentence about confounding, and using the CI formalization.

- While we are at it, let's draw a DAG too.
- Do you believe this? What's a counter-example?
- Can we test whether this assumption, however stated, is valid?

The most important slide

What was the first commandment? How does it apply here?

And what are the assumptions we need to get a causal effect estimate of the type of smoking on death rate?

- Let's state it both in a sentence about confounding, and using the CI formalization.

- While we are at it, let's draw a DAG too.
- Do you believe this? What's a counter-example?
- Can we test whether this assumption, however stated, is valid?
- We'll talk more about assessing the credibility of such assumptions

Estimation

While identification concerns are paramount, lets talk about estimation concerns for a moment.

- Did we need any model? Did we estimate any parameters?

Estimation

While identification concerns are paramount, lets talk about estimation concerns for a moment.

- Did we need any model? Did we estimate any parameters?
- What is worrying about the sub-classification estimation we did though, estimation-wise?

Estimation

While identification concerns are paramount, lets talk about estimation concerns for a moment.

- Did we need any model? Did we estimate any parameters?
- What is worrying about the sub-classification estimation we did though, estimation-wise?
- More generally, why do you think sub-classification is of limited use in most real applications?

Summary

The SOO/CI assumption is a common strategy.

Summary

The SOO/CI assumption is a common strategy.

- Says to find strata of X in which you have a story for why treatment is randomized.

Summary

The SOO/CI assumption is a common strategy.

- Says to find strata of X in which you have a story for why treatment is randomized.
- Is also a way of saying **no unobserved confounders**.

Summary

The SOO/CI assumption is a common strategy.

- Says to find strata of X in which you have a story for why treatment is randomized.
- Is also a way of saying **no unobserved confounders**.

Remember the First Commandment.

Summary

The SOO/CI assumption is a common strategy.

- Says to find strata of X in which you have a story for why treatment is randomized.
- Is also a way of saying **no unobserved confounders**.

Remember the First Commandment.

Your main job: can you argue that variation in treatment status within strata of X is random? How?

Summary

The SOO/CI assumption is a common strategy.

- Says to find strata of X in which you have a story for why treatment is randomized.
- Is also a way of saying **no unobserved confounders**.

Remember the First Commandment.

Your main job: can you argue that variation in treatment status within strata of X is random? How?

Next up: Same assumptions, different estimators. Matching, weighting, regression.